Build with Gemini 4 Argon: UCL AgentVerse 2026 builder guide

This guide is for student engineers, researchers, designers and first-time builders competing in teams of 3 to 5 at AgentVerse 2026, hosted by UCL Artificial Intelligence Society with Google DeepMind. This gives you the model toolkit, weekend pacing plan, runnable code patterns and safety boundaries to ship a working application or autonomous agent in two days.

Gemini 4 Argon combines input, reasoning and output in one shared 1 million token context window. It works longer on its own across full codebases, videos and dense research sets, and pairs as a lead planner and verifier with Antigravity Managed Agents and companion models for live voice, vision, 4K visuals, video, music and local execution.

Google provides onsite Gemini API credits for every builder across all 4 official tracks: Robotics, Finance, Biotech and Creative. Across the weekend, teams compete for Google DeepMind's Best Use of Gemini prize alongside track cash prizes, extra Google Cloud and Gemini API credits, Google DeepMind swag, a potential visit to the Google London office, and venture capital introductions through The Foundry, UCL Artificial Intelligence Society's startup incubator.

What makes a strong two-day project

Judges score live working software, not slide decks. Strong AgentVerse projects pass 4 tests:

- solving a concrete problem for a specific person or role you can name, rather than a generic audience
- doing work that needed Gemini 4 Argon or multimodal orchestration with a verification loop, such as multi-file coding, sandboxed test verification, 1 million token synthesis, or 3D physics simulation
- keeping people in control whenever an action affects external systems, money, clinical care or physical hardware
- running live with real inputs in under 3 minutes during Sunday afternoon judging, backed by a public code repository written during the hackathon

How to pace your two-day weekend

AgentVerse runs from Saturday morning at 9am to Sunday evening at 6pm. Use the pacing plan below so your team proves the core loop on Saturday and protects two calm hours before Sunday's submission deadline.

Weekend phase	Focus	What your team completes
Saturday morning kickoff	Problem and first API call	Claim your onsite Gemini API credits at the Google DeepMind booth, hear the official track problem statements at the opening ceremony, choose one named user workflow, and run your first Interactions API call.

Weekend phase	Focus	What your team completes
Saturday afternoon core loop	Brain and verification gate	Build your core Gemini 4 Argon reasoning and verification loop first, such as a Python risk check, an Antigravity test suite, line-cited protocol rules, or a 3D simulation check.
Saturday evening senses and rest	Multimodal layer and checkpoint	Add live camera, expressive speech, 4K assets or adaptive audio once the core loop works, push a clean working checkpoint to GitHub, and get proper sleep.
Sunday morning feature freeze	Polish and live demo rehearsal	Freeze new features at least 2 hours before the submission deadline, test edge cases with fresh inputs, confirm no API keys sit in your public repository, and rehearse a live demo under 3 minutes.

Your toolkit: Gemini 4 Argon, Managed Agents and companion models

Pick each model or managed agent by the specific job it performs in your application. Most strong teams use Gemini 4 Argon as the central brain and verifier while delegating live camera and voice, parallel subtasks, generative media or offline execution to companion models.

Job in your app	Model or agent	When to use it
Lead reasoning, coding, 3D and orchestration	Gemini 4 Argon	Multi-file software engineering, security patching, 1 million token synthesis, agentic video analysis, Three.js and WebGPU scenes, and multi-agent orchestration.
Autonomous Linux sandbox execution	Antigravity Managed Agent	Starts a Google-hosted Linux sandbox in one API call where Gemini 4 Argon writes code, runs tests, manages files, calls Model Context Protocol tools and searches the web.
Fast parallel subagents and tool calls	Gemini 3.8 Flash	High-speed model for parallel leaf tasks, repository exploration, test generation and low-latency tool calls coordinated by Gemini 4 Argon.
High-frequency routing and checks	Gemini 3.5 Flash-Lite	Lowest-latency intent classification, request routing and lightweight schema validation.
Cited multi-step web research	Deep Research agent	Runs multi-step background web and scientific literature research that returns a structured, cited report.
Real-time voice and camera sessions	Gemini 3.8 Live	Two-way low-latency voice and live camera sessions over WebSockets with mid-sentence interruptions, non-blocking tool calls and an extended thinking option.
Live transcription and translation	Gemini 3.5 Transcribe and Live Translate	Streaming speech-to-text captions, recorded audio transcription with speaker labels, and continuous speech-to-speech translation across more than 70 languages.
Expressive speech and custom voices	Gemini 3.8 Flash TTS	Studio-quality speech in 130 languages using natural-language voice design or consent-verified voice replication, plus a Lite model for faster output.
4K images, UI assets and textures	Nano Banana 2.1	Generates and edits consistent characters, UI sprites, brand typography and 4K textures with Search grounding and aspect ratios up to 8 to 1, plus a Lite model for sub-second drafts.
Video generation with native audio	Veo and Gemini Omni Flash	Use Gemini Omni Flash on the Gemini API to generate and edit video with synchronized sound from text, images, audio or video, or Veo 3.1 on Vertex AI for 4K cinematic clips.
Songs, vocals and adaptive loops	Lyria 3.5	Produces full 44.1 kHz stereo songs with vocals and timed lyrics, or 30-second instrumental loops for interactive scenes and brand soundtracks.

Job in your app	Model or agent	When to use it
Private or offline open models	Gemma 4	Open-weight models on the Interactions API, Kaggle and Hugging Face, including compact E2B and E4B sizes that run locally on laptops, phones and robots with native audio input.

Copy the Python dictionary below into your project configuration so your team has the exact API model and agent strings ready to use.

```python
from google import genai

MODEL_IDS = {
    # Lead frontier reasoning, coding, 3D/WebGPU, and orchestration
    "lead_reasoning": "gemini-4-argon",
    # Hosted Linux sandbox and autonomous managed agents
    "managed_sandbox": "antigravity-preview-09-2026",
    "deep_research": "deep-research-preview-04-2026",
    # Fast parallel subagents and low-latency routing
    "fast_subagent": "gemini-3.8-flash",
    "fast_router": "gemini-3.5-flash-lite",
    # Real-time bidirectional voice, camera, transcription, and translation
    "live_voice_vision": "gemini-3.8-live",
    "live_extended_thinking": "gemini-3.8-live-extended-thinking",
    "transcribe_live": "gemini-3.5-transcribe-live",
    "transcribe_batch": "gemini-3.5-transcribe",
    "live_translate": "gemini-3.5-live-translate-preview",
    "tts_studio": "gemini-3.8-flash-tts",
    "tts_fast": "gemini-3.8-flash-lite-tts",
    # Generative images, video with audio, and music
    "image_4k": "gemini-nano-banana-2.1",
    "image_fast": "gemini-3.1-flash-lite-image",
    "video_audio": "gemini-omni-1.1-flash",
    "video_cinematic_vertex": "veo-3.1-generate-001",
    "music_full_song": "lyria-3.5",
    "music_loop": "lyria-3-clip-preview",
    # Open-weight model on the Interactions API
    "open_model": "gemma-4-31b-it",
}

client = genai.Client()
```

Get started in 15 minutes

1. Claim your onsite hackathon account or Gemini API credits from the Google DeepMind booth during Saturday morning check-in.
2. Follow the hackathon account setup guide to sign in, open Google AI Studio and generate your Gemini API key.

3. Install the official Gemini API skills in your coding agent so it writes current Gemini 4 Argon and Interactions API code, or start prototyping in Google AI Studio Build, AI Studio Live and Google Antigravity.
4. Install the Google GenAI SDK version 2.3.0 or later for Python or TypeScript and run your first Gemini 4 Argon call on the Interactions API.
5. Visit the Google DeepMind booth throughout Saturday and Sunday whenever you want help shaping your architecture, verification loop or live demo.

Install the official Gemini API skills in your coding agent

Coding assistants trained on older repositories often suggest deprecated SDK packages or retired parameters. Run these commands in your terminal first so your coding agent uses current Gemini 4 Argon, Interactions API and Managed Agents patterns.

```shell
pip install -U "google-genai>=2.3.0"
npm install @google/genai@latest

npx skills add google-gemini/gemini-skills --skill gemini-api-dev
npx skills add google-gemini/gemini-skills --skill gemini-live-api-dev
agy plugin install https://github.com/google-gemini/gemini-skills
```

What to know when calling Gemini 4 Argon

The Interactions API is the unified endpoint for both Gemini models and Managed Agents. It stores multi-turn state on the server through previous interaction identifiers and supports asynchronous background execution for long-running builds.

Control reasoning depth with thinking level

Gemini 4 Argon uses the thinking level parameter to set how much reasoning effort the model spends before responding:

- the medium level is the default and balances speed, cost and first-pass accuracy for interactive coding and tool calls
- the high level spends deeper reasoning effort on multi-file debugging, security patching, mathematical proofs, financial risk models and upfront system design
- the low level minimizes latency for fast routing, classification and targeted extraction

```python
from google import genai

client = genai.Client()
```

```
interaction = client.interactions.create(
    model="gemini-4-argon",
    input=(
        "Build a single-file HTML synthesizer and 16-step sequencer using "
        "the Web Audio API with a tempo slider and live waveform canvas."
    ),
    generation_config={
        "thinking_level": "medium",  # "low" | "medium" (default) | "high"
    },
)
print(interaction.output_text)
```

Use the 1 million token window and background execution

Earlier models capped combined thinking and response output at 64,000 tokens. In Gemini 4 Argon, input, reasoning and output share one 1 million token window:

- the default output limit on the Interactions API is 65,536 tokens for both streaming and non-streaming calls with no extra configuration
- to generate past 65,536 tokens up to the full 1 million token window minus your input, enable background execution and set maximum output tokens above 65,536

Python

```
import time
from google import genai

client = genai.Client()

interaction = client.interactions.create(
    model="gemini-4-argon",
    input=(
        "Design and implement a complete multi-module physics simulation "
        "with deterministic collision checks and a full pytest suite."
    ),
    background=True,
    generation_config={
        "thinking_level": "high",
        "max_output_tokens": 262144,
    },
)

while interaction.status in ("in_progress", "queued"):
    time.sleep(5)
    interaction = client.interactions.get(id=interaction.id)
```

```
print(interaction.output_text)
```

Run agentic video analysis and remove legacy parameters

Gemini 4 Argon supports agentic video processing, which inspects transcripts, audio tracks and high-resolution frames on demand across long recordings while using up to 88% fewer tokens than static frame sampling.

Python

```
from google import genai

client = genai.Client()

video_file = client.files.upload(file="robot_bench_trial.mp4")
telemetry_csv = client.files.upload(file="joint_torque_log.csv")

analysis = client.interactions.create(
    model="gemini-4-argon",
    input=[
        {"type": "video", "uri": video_file.uri, "processing": "agentic"},
        telemetry_csv,
        (
            "Scrub the bench trial video alongside the joint torque CSV. "
            "Locate the exact timestamp where gripper alignment drifts, "
            "identify the mechanical cause, and output a patched calibration
config."
        ),
    ],
    generation_config={"thinking_level": "high"},
)
print(analysis.output_text)
```

If you are adapting code written for Gemini 2.5 or Gemini 3 models, apply these 4 cleanup rules:

- remove the deprecated thinking budget parameter and pass thinking level instead
- remove custom temperature, top-p and top-k sampling parameters, which Gemini 4 Argon ignores with a deprecation warning
- remove manual context-caching setup calls, because implicit context caching runs automatically at a 95% discount on repeated input prefixes, alongside introductory pricing of $2 per 1 million input tokens and $10 per 1 million output tokens through 30 November 2026
- raise HTTP client timeouts to 15 to 20 minutes on synchronous high-thinking calls, or run long tasks asynchronously in the background

Run Gemini 4 Argon inside Antigravity Managed Agents

Pass Gemini 4 Argon to the Antigravity Managed Agent when your task needs real code execution. Google starts an isolated Linux sandbox where Gemini 4 Argon plans, writes code, runs compilers or headless browser tests, inspects errors and repairs its own work in one API call.

```python
from google import genai

client = genai.Client()

interaction = client.interactions.create(
    agent="antigravity-preview-09-2026",
    agent_config={
        "type": "antigravity",
        "model": "gemini-4-argon",
    },
    input=(
        "Clone our starter repository, write a failing pytest reproduction "
        "for the order-book rounding bug, patch the calculation module, and "
        "iterate until the full test suite passes with zero warnings."
    ),
    environment="remote",
)
print(interaction.output_text)
```

When your agent needs external API tokens such as GitHub or data-provider keys, register them with the Credentials API and set a sandbox network allowlist so secrets are injected at the egress proxy and never enter the model context window.

Domain constraints and what we left out

Follow these boundaries all weekend to keep your system minimal, reliable and eligible for judging:

- do not submit a basic PDF chat wrapper or single-document question-and-answer bot with no tool execution, planning or verification loop
- do not show mocked screens or hardcoded JSON payloads that pretend to call Gemini live
- do not build unverified medical diagnosis, clinical prescribing or autonomous financial trading tools without deterministic verification checks and human approval gates
- do not demo application code written before Saturday morning's kickoff, though open-source libraries, public datasets and starter templates are welcome

- do not commit Gemini API keys or third-party credentials to public GitHub repositories, or paste raw secrets into prompts instead of using environment variables and the Credentials API
- do not hand-roll custom container sandboxes or client retry state machines when the Interactions API and Antigravity Managed Agents already handle server-side state and hosted Linux execution

This guide intentionally leaves out boilerplate login screens, user account management, database setup and generic web deployment scaffolding. Spend your weekend hours on the specific user problem, the Gemini 4 Argon verification loop and a live Sunday demo that works on real inputs.

Resources and support

Use these official guides and starter repositories while you build:

- read the Gemini 4 Argon developer guide for complete API parameters and examples
- generate interactive web apps and 3D prototypes in Google AI Studio Build
- test real-time camera and voice barge-in sessions in AI Studio Live and design custom voices in Speech Studio
- run autonomous multi-file builds and sandboxes in Google Antigravity
- add the official Gemini API skills to your coding agent so it writes current SDK version 2.3.0 code
- browse the Gemini API documentation and inspect WebSocket voice and vision starters in the Live API starter examples

The Google DeepMind team is at the booth all weekend to review your architecture, help debug API calls and watch practice runs of your demo.

Tracks and build blueprints

UCL Artificial Intelligence Society reveals the official track problem statements live at Saturday morning's opening ceremony. Once those prompts are announced, use the blueprints below to combine Gemini 4 Argon, Antigravity Managed Agents and companion models across all 4 official AgentVerse tracks and contend for Google DeepMind's Best Use of Gemini prize.

Track 1: Robotics

Supported by NVIDIA and UCL Robotics Society, the Robotics track challenges teams to move agents out of browser chat windows so they perceive, reason and act safely in simulated or physical environments.

What Gemini 4 Argon adds to Robotics

Gemini 4 Argon gives robotics teams three practical capabilities in a two-day build:

- generating complete MuJoCo scenes, URDF robot descriptions, inverse kinematics scripts and Three.js visualizers across a shared 1 million token context window
- running closed-loop visual self-critique where an Antigravity sandbox renders multi-angle camera views and repairs collisions or floating geometry automatically
- pairing cloud safety verification with Gemini 3.8 Live for 30 frames-per-second webcam and voice control, or local Gemma 4 E2B and E4B models when an edge controller has no network link

Two-day build ideas for Robotics

Adapt any of these starter architectures once the Saturday Robotics prompt is revealed:

- a bench-photo-to-simulation digital twin that photographs a tabletop robot arm or workcell, builds a physics-verified 3D MuJoCo or Three.js scene with Gemini 4 Argon, and inspects four-angle screenshots in a loop until every trajectory clears obstacles safely
- a tabletop webcam and live voice drone or rover fleet coordinator where Gemini 3.8 Live watches a physical arena, handles spoken mission changes mid-sentence, and calls Gemini 4 Argon to recompute collision-free paths across multiple simulated vehicles
- an offline-first Gemma 4 edge planner for a mobile robot or assistive wearable that processes local audio and obstacle cues on a laptop, while verifying multi-step recovery plans against Gemini 4 Argon safety constraints whenever connectivity returns

Track 2: Finance

Supported by Jane Street, the Finance track asks for transparent risk, market and everyday financial tools that show their reasoning, handle uncertainty honestly and earn a professional's trust near real money.

What Gemini 4 Argon adds to Finance

Gemini 4 Argon strengthens financial decision engines in three ways:

- holding entire regulatory filings, transaction ledgers, contract PDFs and webcast videos together in one 1 million token window with automatic 95% implicit caching on follow-up queries
- combining high-thinking scenario reasoning in Gemini 4 Argon with fast parallel document extraction across Gemini 3.8 Flash subagents
- proving every calculation, risk limit and accounting balance inside an Antigravity Python or linear-programming solver before presenting a human approval gate

Two-day build ideas for Finance

Use these architectures to pair deep reasoning with exact mathematical proof:

- an earnings video and filing cross-examiner that scrubs executive webcast video using agentic video processing alongside quarterly tables in Gemini 4 Argon, runs Python verification scripts to check every stated margin and cash-flow figure, and cites exact timestamps and lines
- a flash-crash portfolio stress tester that simulates limit-order-book shocks at high thinking level, gates every proposed hedge through a deterministic Python or linear-programming solver so invalid dispatches cannot pass, and pauses at a human circuit-breaker screen
- a multi-source insurance claims or freight discrepancy auditor that cross-examines incident footage, shipping manifests, public vehicle or trade APIs, and contract terms using exact decimal arithmetic to produce an auditable settlement report

Track 3: Biotech from lab to patient

The Biotech track at AgentVerse spans the full path from early drug and genomic discovery to clinical trial matching and bedside clinician support. Always build on public or synthetic datasets, cite your sources clearly and keep clinicians in control of care decisions.

What Gemini 4 Argon adds to Biotech

Gemini 4 Argon helps teams bridge scientific discovery and clinical workflows:

- synthesizing dozens of scientific papers, assay logs, genomic tables and clinical trial protocols in one 1 million token window without brittle document chunking

- running reproducible Python or PyTorch hypothesis checks inside an Antigravity sandbox alongside cited literature reports from the Deep Research agent
- turning dense clinical protocols into clear bedside conversations across more than 70 languages with Gemini 3.8 Live, Live Translate and Gemini 3.8 Flash TTS

Two-day build ideas for Biotech

Consider these patterns across wet-lab discovery and patient care:

- an autonomous literature and assay hypothesis verifier in Antigravity that reads public target papers and prior assay logs in Gemini 4 Argon, writes and executes Python or PyTorch validation benchmarks, and ranks candidate designs by measured results
- a cited clinical trial protocol matcher over public trial registries and synthetic patient profiles that checks every inclusion and exclusion rule with exact line citations, flags missing lab values for clinician approval, and generates a clear voice briefing in 70 languages
- an accessible multimodal medication and care companion where patients or carers point a camera at public sample medicine packaging via Gemini 3.8 Live, verify dosing schedules deterministically in Python, and hear calm spoken guidance with a clinician escalation step

Track 4: Creative and Brandathon

Supported by Canva, the Creative track includes UCL's Brandathon challenge and is open to non-technical builders, designers and storytellers from every degree background. You can direct working web software, 3D scenes and generative media in natural language.

What Gemini 4 Argon and generative media add to Creative

Combining Gemini 4 Argon with Google DeepMind's media models turns static concepts into interactive experiences:

- building complete single-file web apps, interactive layouts and 60 frames-per-second Three.js or WebGPU worlds from natural-language prompts in Google AI Studio Build
- generating consistent 4K logos, campaign posters and crisp typography with Nano Banana 2.1, original songs and sonic branding with Lyria 3.5, and synchronized launch videos with Gemini Omni Flash or Veo
- giving your brand or story world a live two-way voice presence through Gemini 3.8 Live so judges can speak with it directly on stage

Two-day build ideas for Creative

Any team can ship one of these live creative experiences in a weekend:

- a living Brandathon studio for a charity, campus movement or London startup that pairs Gemini 4 Argon brand strategy with 4K Nano Banana 2.1 campaign posters, a Lyria 3.5

sonic anthem, a Veo and Gemini Omni Flash launch trailer, and a live Gemini 3.8 Live voice persona
- a photo-to-playable 3D browser world that turns smartphone photos of Bloomsbury architecture or tabletop props into stylized Nano Banana 2.1 textures and a self-critiqued Three.js or WebGPU world running in Chrome
- an interactive voice-driven story world where players talk freely with unscripted characters powered by Gemini 3.8 Live and Gemini 3.8 Flash TTS while Gemini 4 Argon updates plot state, visual scenes and adaptive Lyria 3.5 music in real time

How to contend for Best Use of Gemini

Because Google DeepMind is Exclusive AI Partner across all 4 tracks, every project at AgentVerse can contend for the Best Use of Gemini prize alongside its track award. Winning submissions combine 3 elements:

- using 2 or more modalities in one connected loop, such as live camera with duplex voice, video evidence with 3D physics, or 4K brand visuals with custom speech and music
- verifying Gemini 4 Argon outputs through a real check, such as a Python math or physics solver, an Antigravity test suite, line-by-line rule citations, or multi-angle screenshot critique
- delivering an interactive experience that could never happen inside a plain text box, while keeping people in control of real-world actions

Sunday pre-submission checklist

1. Freeze new features at least 2 hours before Sunday's submission deadline so your team has dedicated time for testing and demo polish.
2. Run your live application end to end with fresh, realistic inputs and remove any brittle placeholder buttons or dead links.
3. Audit your public GitHub repository to confirm that no Gemini API keys or third-party secrets appear in source files or commit history.
4. Write a concise README stating the named user problem, the exact Gemini 4 Argon and companion model identifiers used, and how your verification loop works.
5. Rehearse a live demo under 3 minutes that opens directly on your working software within 30 seconds, triggers your multimodal and verification loop live, and keeps a local screen recording ready as a Wi-Fi backup.